

smart camera that can stream data to the cloud will have a different connectivity profile than a smart energy meter.

Despite this, existing operator-based cellular technologies are the backbone for enabling a lot of these connected IoT products. A mature ecosystem backed by global standards currently provides seamless interoperability for these devices all over the world.

The rise of LTE for the IoT

Long Term Evolution (LTE) is a 4G wireless communications standard developed by 3rd Generation Partnership Project (3GPP), a standards organisation that develops protocols for mobile telephony.

LTE provides a stable base for many connected devices. For high-performance IoT, there is LTE CAT-1, and for higher throughput data (for example, connected cars and connected cameras), there is LTE CAT-4.

In addition, there are a couple of complimentary LTE IoT technologies, namely, EMT-CAT-M1 and NB-IOT CAT-NB1, that provide different kinds of connectivity profiles and throughputs for different use-cases.

CAT-M1 is ideally suited for wearables, health monitors and security systems because of the increased amount of the data these consume. On the other hand, CAT-NB1 requires very little data—about 10kbps or so—to connect such things as parking meters or smart meters. Most devices with these features are extremely simple, have low complexity, single receiver antennae and operate in half duplex mode. These also support a long battery life as these have far more efficient power-saver modes.

LTE IoT also provides deeper coverage with extra link budget up to 20dB. In addition, simplified signalling schemes allow a lot more devices to connect in a single cell to an LTE IoT network.

Strong global momentum of LTE IoT

LTE IoT is the first step towards a truly connected IIoT base. There are more than fifty commercial CAT-M1 and/or CAT-NB1 networks in over thirty countries. These are in various stages of deployment—either deployed or in

the process of testing.

In India, a lot of testing is happening on CAT-NB1; whereas, countries like the US and China have a mix of both. This shows that there is a global scale for LTE IoT infrastructure and ecosystem.

There is a requirement for both cloud and Edge to achieve efficient distribution of intelligence required for massive 5G.

Despite all this, it is still all about the cloud. That means that billions of IoT devices are still feeding data to the cloud, and are letting the cloud make intelligent choices. The cloud decides which sensor to turn off, which machine to turn on and which meter to pole for data, and so on.

Artificial intelligence (AI) is still in the cloud connected to a lot of devices, and that is where it takes decisions. To scale to billions or even trillions of devices, AI in the cloud needs to go to Edge. There needs to be a shift in the centre of gravity; **devices need to be more intelligent on the Edge, rather than relying on the cloud all the time.**

This is because, as devices become more intelligent, the model of sending data from one cloud to the next just does not scale—the capacity is not there. Hence, the concept of Wireless Edge is being introduced in 5G.

Use-cases of Wireless Edge

Some industrial use-cases of Wireless Edge encompass construction, ports and shipping, mines, refineries, manufacturing and warehousing. All of these require a higher degree of reliability, connectivity and faster decision-making. In ten to fifteen years, the size of this market will be over five trillion dollars. There is immense scale if LTE IoT technology can evolve to fit into this massive ecosystem of the IIoT. These sectors will drive the major chunk of demand for Wireless Edge.

Various stages of Industry X.0 were mechanisation, electrification, digitisation and connectivity. The fourth industrial revolution, that is, Industry 4.0, is driven by Big Data and machine learning, which are enabled by connectivity. If 5G is added to Industry 4.0, then it will reach the next level. Then, we can not only reconfigure factories, we can also have automated, guided vehicles

enabling mobile robots. All this can be controlled by a combination of 5G and Industry 4.0.

On an industrial floor, there are robots, workers, analytics on the Edge, different kinds of sensors, security cameras and recording machines. How can we bring these devices together and automate them? We cannot put Ethernet on all of them or wire them, because all have to be reconfigurable and moving all the time.

Besides, each of these has its own set of requirements. For example, sensors need very low latency monitoring. Security cameras not only need data throughput but also an extremely high level of AI and machine learning to know what is happening on the floor. The head-mounted display needs to provide a worker with 360-degree view of what he is working on. There are many such examples. And all of this is possible in real time with 5G.

Essentially, 5G paradigm for the IIoT is a combination of security, connectivity and compute.

LTE IoT evolving into 5G for IIoT

Today, there is NB-1 that connects low-data devices, and a step higher than this is CAT M-1—these fit into 5G and 5G NR (New Radio). Starting release 16, 3GPP has introduced the concept of 5G NR, which dovetails into migration of taking LTE IoT into 5G.

How do we move all this intelligence that is on the cloud to the Edge? Should it all be on the cloud, or on the Edge?

Putting everything on the cloud is inefficient. Also putting everything on the Edge may not be the right option. It should be the combination of both. By using both, we can achieve an efficient distribution of intelligence that is required for massive 5G.

On-Edge devices, which use low latency or very minute packets of data, can reside on the Edge. Whereas, Edge cloud can cover products such as XR, which need more data.

Advantages of 5G

5G is not restricted to one band; it is fairly flexible and operates in multiple bands. It can operate in low